Compactness through finite intersections

Theorem 2.2.2: Compactness by closed sets

All levels · read along the `refines` order

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Contents

2.2.2 Compactness by closed sets **1**

2.2.2.1 Proof of Theorem 2.2.2 1

Outside this export **1**

Theorem 2.2.2 (Compactness by closed sets)**.** *X is compact if and only if every family $(F_i)_{i\in I}$ of closed subsets of X with the finite intersection property has $\bigcap_{i\in I} F_i \neq \emptyset$.*

Proof 2.2.2.1. Of Compactness by closed sets (Theorem 2.2.2).

Both directions are the contrapositive of each other once the complements are taken: Lemma 2.2.2.1.1 exchanges the two conditions, and Lemma 2.2.2.1.2 says what a finite subcover becomes.

USES → Finite intersection property (Definition 2.1.2*)
Where the finite intersection property enters. Only in Lemma 2.2.2.1.2: the property is the statement that no finite subfamily has empty intersection, word for word.

Lemma 2.2.2.1.1 (Complements)**.** *For a family $(F_i)_{i\in I}$ of closed sets put $U_i = X \setminus F_i$. Then*

$$\bigcap_{i\in I} F_i = \emptyset \iff \bigcup_{i\in I} U_i = X,$$

so (U_i) is an open cover exactly when the F_i have no common point.

Lemma 2.2.2.1.2 (Finite subfamilies)**.** *For $J \subseteq I$ finite, $(U_j)_{j\in J}$ covers X if and only if $\bigcap_{j\in J} F_j = \emptyset$. Hence (U_i) has no finite subcover if and only if (F_i) has the finite intersection property.*

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References marked * are to parts of the blueprint this document leaves out:

Definition 2.1.2 Finite intersection property, in 2 Covers and compact spaces (`def-fip`)